Polona Durcik

Chapman University
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Research interests: Harmonic analysis and its connections with ergodic theory and arithmetic combinatorics.

Employment

2020/08 – present	Assistant professor, Chapman University, Orange, CA
2017/09 – 2020/07	Harry Bateman Instructor, California Institute of Technology, Pasadena, CA

Education

2017/07	Ph.D. Mathematics, University of Bonn, Germany Advisor: Christoph Thiele
2014/02	M.Sc. Mathematics, University of Bonn, Germany
2011/09	B.Sc. Mathematics, University of Ljubljana, Slovenia

Publications and preprints

- (with M. Fraccaroli, J. Roos), *A curved three-dimensional simplex Hilbert transform*, submitted (2026).
- (with M. Fraccaroli, J. Roos), *A weak Hellinger inequality for noisy Boolean channels*, submitted (2026).
- (with F. van Doorn, J. Roos, L. Slavíková, C. Thiele), *A blueprint for the formalization of norm-variation of multiple ergodic averages for commuting transformations* (2026).
- (with L. Becker) *The shifted bilinear Hilbert transform*, submitted (2026).
- (with P. Ivanisvili, J. Roos, X. Xie) *Sharp isoperimetric inequalities on the Hamming cube II: The critical exponent*, submitted (2026).
- (with K. A. Škreb) *Norm-variation of cubic ergodic averages*, J. Math. Anal. Appl. 562 (2026), no. 2, Article 130689.
- (with L. Becker, F. Y-H. Lin) *On trilinear singular Brascamp-Lieb integrals*, J. London Math. Soc., 114 (2026), no. 4, e70729.
- (with P. Ivanisvili, J. Roos) *Sharp isoperimetric inequalities on the Hamming cube near the critical exponent*, To appear in Adv. Math. (2024).
- (with K. Domelevo, V. Fragkiadaki, O. Klein, D. Oliveira e Silva, L. Slavíková, B. Wróbel) *Dimension-free inequalities for low degree functions on the Hamming cube*, Studia Math. 280 (2025), 161-173.
- (with L. Slavíková, C. Thiele) *Norm-variation of triple ergodic averages for commuting transformations*, Analysis & PDE. 19 (2026), no. 3, 539-586.
- (with V. Kovač, M. Stipčić) *A strong-type Furstenberg-Sárközy theorem for sets of positive measure*, J. Geom. Anal. 33 (2023), issue 8, no. 255, 16 pp.

- (with J. Roos) *A new proof of an inequality of Bourgain*, Math. Res. Lett. 31 (2024), no. 4, 1047-1060.
- (with M. Stipčić) *Quantitative bounds for products of simplices in subsets of the unit cube*, Isr. J. Math. 267 (2025), 135-156.
- (with L. Slavíková, C. Thiele) *Local bounds for singular Brascamp-Lieb forms with cubical structure*, Math. Z. 302 (2022), 2375-2405.
- (with R. Greenfeld, A. Iseli, A. Jamneshan, J. Madrid) *An uncountable ergodic Roth theorem and applications*, Discrete Contin. Dyn. Syst., 42(11) (2022), 5509-5540.
- (with M. Christ, V. Kovač, J. Roos) *Pointwise convergence of certain continuous-time double ergodic averages*, Ergodic Theory Dynam. Systems. 42 (2022), no. 7, 2270-2280.
- (with M. Christ, J. Roos) *Trilinear smoothing inequalities and a variant of the triangular Hilbert transform*, Adv. Math. 390 (2021), Paper No. 107863.
- (with F. Bernicot) *Boundedness of some multi-parameter fiber-wise multiplier operators*, Indiana Univ. Math. J. (2020).
- (with V. Kovač) *A Szemerédi-type theorem for subsets of the unit cube*, Analysis & PDE. 15 (2022), no. 2, 507-549.
- (with L. Cladek, B. Krause, J. Madrid) *Directional maximal function along the primes*, Publ. Mat. 65 (2021), no. 2, 841-858.
- (with C. Thiele) *Singular Brascamp-Lieb: a survey*, Geometric Aspects of Harmonic Analysis, Vol. 45, Springer INdAM Series, Springer International Publishing, 2021.
- (with J. Roos) *Averages of simplex Hilbert transforms*, Proc. Amer. Math. Soc. 149 (2021), no. 2, 633-647.
- (with V. Kovač) *Boxes, extended boxes, and sets of positive upper density in the Euclidean space*, Math. Proc. Cambridge Philos. Soc., 171 (2021), no. 3, 481-501.
- (with C. Thiele) *Singular Brascamp-Lieb inequalities with cubical structure*, Bull. London Math. Soc. 52 (2020), no. 2, 283-298.
- (with S. Guo, J. Roos) *A polynomial Roth theorem on the real line*, Trans. Amer. Math. Soc. 371 (2019), no. 10, 6973-6993.
- (with V. Kovač, L. Rimanić) *On side lengths of corners in positive density subsets of the Euclidean space*, Int. Math. Res. Not. 2018, no. 22, 6844-6869.
- (with V. Kovač, C. Thiele) *Power-type cancellation for the simplex Hilbert transform*, J. Anal. Math. 139 (2019), 67-82.
- (with V. Kovač, K. A. Škreb, C. Thiele) *Norm-variation of ergodic averages with respect to two commuting transformations*, Ergodic Theory Dynam. Systems 39 (2019), no. 3, 658-688.
- *L^p estimates for a singular entangled quadrilinear form*, Trans. Amer. Math. Soc. 369 (2017), no. 10, 6935-6951.
- *An L^4 estimate for a singular entangled quadrilinear form*, Math. Res. Lett. 22 (2015), no. 5, 1317-1332.

Funding

- NSF DMS-2554859, 2026–2029.
- Simons Foundation MPS-TSM-00013943, 2025–2030.
- NSF DMS-2154356, 2022–2025.
- AIM SQuaRE; with I. Holmes Fay, P. Ivanisvili, J. Roos, A. Volberg, 2021–2024.
- AMS–Simons travel grant, 2018–2020.
- Hausdorff scholarship; doctoral scholarship by Bonn International Graduate School of Mathematics, 2014–2017.
- Zois scholarship; Slovenian national merit scholarship, 2008–2014.

Talks

Invited talks (colloquia, seminars, and conferences)

2026/10 Pure Mathematics Seminar, California State University Fullerton, CA.
2026/10 AMS Meeting, session 'Harmonic Analysis and PDEs', George Washington University, Washington, DC.
2026/07 Philadelphia Harmonic Analysis and Differential Equations, 2026 ICM Satellite Conference, PA.
2026/07 Graduate Seminar on Advanced Topics in PDE, Univ. of Bonn, Germany.
2026/06 Geometric Harmonic Analysis in the Discrete and Continuous Settings, University of Granada (BIRS-IMAG), Spain.
2026/06 Nonlinear Analysis, Function Spaces and Applications Spring School (NAFSA 13), Prague, Czech Republic.
2026/04 AMS Meeting, session 'Harmonic Analysis', North Dakota State University, Fargo, ND.
2026/03 AMS Meeting, session 'Geometric and Functional Inequalities: Sharp Constants, Extremals, Stability and Their Applications', Boston, MA.
2026/02 Seminar on Analysis and Stochastic Analysis, Auburn University, AL.
2025/11 Schmid Science Forum, Chapman U., CA.
2025/10 Harmonic Analysis Workshop, Madison, WI.
2025/08 SUMIRFAS 2025, Texas A&M University, TX.
2025/04 Nonlinear Analysis Seminar, online at National Taiwan Normal University.
2024/12 NCTS workshop on Harmonic Analysis, National Taiwan University, Taipei, Taiwan.
2024/10 Workshop 'Analysis and Geometry on Discrete Spaces', HIM, Bonn, Germany.
2024/08 Workshop 'Harmonic Analysis in Australia', ANU, Canberra, Australia.
2024/07 Joint meeting AMS-UMI, session 'Harmonic Analysis and Geometric Measure theory', Università degli Studi di Palermo, Italy.
2024/07 Hausdorff school 'Uniformity and Stability of Oscillatory Integrals'. Bonn, Germany.
2024/06 On the Interface of Geometric Measure Theory and Harmonic Analysis, Banff, Canada.
2024/05 Madison Lectures in Fourier Analysis 2024, Madison, WI.
2024/04 Ghent Methusalem Junior Seminar, online.
2024/02 Friedrich-Alexander University of Erlangen-Nürnberg, Germany.
2023/11 Analysis Seminar, UC Irvine, CA.
2023/03 AMS meeting, session 'Discrete Analysis', Georgia Tech, GA.
2023/03 AMS meeting, session 'Harmonic Analysis', Georgia Tech, GA.
2023/01 AMS Joint Mathematics Meeting, session 'Distance problems in continuous, discrete and finite field settings', Boston, MA.
2022/11 Workshop 'Extremal Problems in Harmonic Analysis, Convexity, and Bellman Functions', ICERM, RI.
2022/11 Harmonic Analysis and Fractal Geometry Seminar, UBC, Canada, online.
2022/09 RTG Harmonic Analysis Workshop, UW Madison, WI.

2022/09 Analysis seminar, Brown University, RI.

2022/07 Real Analysis, Harmonic Analysis, and Applications, Oberwolfach, Germany.

2022/06 Workshop on Operator Theory with an Eye on Linear Systems and Hypercomplex Analysis, Chapman University, CA.

2022/03 Ergodic theory and analysis seminar, Rutgers University, NJ, online.

2022/03 AMS meeting, session 'Crossroads: ergodic theory, harmonic analysis, and combinatorics', Tufts University, MA, online.

2022/01 Seminar, HIM trimester program 'Interactions between Geometric measure theory, Singular integrals, and PDE'. Bonn, Germany.

2022/01 Virtual Harmonic Analysis Seminar, UK, online.

2021/10 AMS meeting, session 'Harmonic Analysis', University of New Mexico, Albuquerque, NM, online.

2021/09 Conference 'Baby Steps Beyond the Horizon: Mathematics for Students'. Bedłewo, Poland, two online talks.

2021/06 8th European Congress of Mathematics, minisymposium 'Harmonic Analysis and Partial Differential Equations', Portorož, Slovenia, online.

2021/06 International Prague seminar on function spaces, Charles University Prague, Czech Republic, online.

2021/03 HA-GMT-PDE seminar, University of Minnesotta and University of Missouri, online.

2021/02 Caltech/UCLA joint Analysis seminar, online.

2020/11 The Second Mid-Atlantic Analysis Meeting, seminar, online.

2020/10 Analysis and Geometry seminar, Chapman University, CA, online.

2020/09 Analysis seminar, University of Wisconsin-Madison, WI, online.

2020/09 Colloquium, University of Massachusetts-Lowell, MA, online.

2020/08 8th Pacific Rim Conference in Mathematics, session 'Classical harmonic analysis and combinatorics', Berkeley, CA, online.

2020/02 Colloquium, Oregon State University, OR.

2020/01 Colloquium, University of Alabama, Tuscaloosa, AB.

2020/01 AMS Joint Mathematics Meeting, session 'The Interplay of Harmonic Analysis and Geometry', Denver, CO.

2020/01 Analysis Seminar, UCLA, CA.

2019/12 Colloquium, Stony Brook University, NY.

2019/12 Max Dehn Seminar, University of Utah, UT.

2019/12 Colloquium, University of Utah, UT.

2019/12 Colloquium, Washington University in St. Louis, MO.

2019/11 Colloquium, University of Waterloo, ON, Canada.

2019/11 Analysis seminar, Georgia Tech, GA.

2019/10 Analysis seminar, University of Oregon, OR.

2019/10 Analysis seminar, IU Bloomington, IN.

2019/09 Colloquium, CSU Fullerton, CA.

2019/09 AMS meeting, session 'Recent Developments in Harmonic Analysis', Madison, WI.

2019/09 PANTHA (Purdue Analytic Number Theory and Harmonic Analysis seminar) seminar, Purdue University, IN.

2019/07 Analysis seminar, University of Nantes, France.

2019/07 Analysis seminar, ICTP, Trieste, Italy.

2019/05 RTG Workshop in Fourier Analysis, Madison, WI.

2019/04 Analysis and PDE seminar, USC, CA.

2019/02 Analysis seminar, UC Irvine, CA.

2018/11 Graduate Seminar on Advanced Topics in PDE, Univ. of Bonn, Germany.

2018/10 Analysis seminar, UW Madison, WI.

2018/09 Analysis seminar, Cornell, Ithaca, NY.

2018/05 RTG Workshop in Fourier Analysis, Madison, WI.
2017/11 Analysis Seminar, UNM, Albuquerque, NM.
2017/10 Caltech/UCLA Joint Analysis Seminar, Caltech, CA.
2017/09 Harmonic Analysis and Related Areas, Clay Research Workshop, Oxford, UK.
2017/07 Real Analysis, Harmonic Analysis, and Applications, Oberwolfach, Germany.
2017/06 Workshop on harmonic analysis, The Chinese University of Hong Kong.
2017/05 Seminar for Complex Analysis, University of Ljubljana, Slovenia.
2017/03 Atelier d'Analyse Harmonique, Nantes, France.
2017/02 Harmonic Analysis and Differential equations seminar, UC Berkeley, CA.
2017/01 Analysis and PDE seminar, UCLA, CA.
2017/01 Analysis and PDE seminar, USC, CA.
2017/01 Harmonic Analysis seminar, University of British Columbia, Canada.
2016/12 Graduate Seminar on Advanced Topics in PDE, Univ. of Bonn, Germany.
2016/04 Real Analysis seminar, Paul Sabatier University, Toulouse, France.
2016/02 Probability seminar, University of Zagreb, Croatia.
2015/11 Graduate Seminar on Advanced Topics in PDE, Univ. of Bonn, Germany.

Contributed conference talks and posters

2019/09 Women in Mathematics in SoCal, CSU Channel Islands, CA. (talk)
2016/08 Conference Harmonic Analysis, Complex Analysis, Spectral Theory and all that, Bedłewo, Poland. (talk)
2016/05 Conference in Harmonic Analysis in honor of M. Christ, Madison, WI. (poster)

Selected research visits

- ANU, 'Special year in Harmonic Analysis'. Canberra, Australia. 8/11-8/30/2024.
- ICERM, semester program 'Harmonic Analysis and Convexity'. Providence, RI. 9/7-12/3/2022.
- HIM, trimester programs, Bonn, Germany. 1/10-1/31/2022 and 9/9-12/18/2024.
- MSRI, semester program 'Harmonic Analysis'. Berkeley, CA. 1/18-2/17/2017.
- Several short academic visits (UW Madison, U. Nantes, U. Bonn, U. Zagreb, CU Hong Kong) and other conferences and summer schools attended (gave expository talks), 2013–present.

Teaching

Chapman University

- Math 451 (Complex Analysis), S26
- Math 100 (Learning and Studying Math for Understanding), S25, F25, F26
- CS 660 (Fourier Analysis), S24
- Math 210 (Multivariable Calculus), F23, F26
- Math 491 (Student-Faculty Research and Creative Activity), S22, F23, F26
- Math 440 (Topology), S22, S23, S24
- Math 450 (Real Analysis), F21
- Math 111 (Single Variable Calculus II), S21
- Math 110 (Single Variable Calculus I), F20, F21, S22, S23

Caltech

- Ma 110 (Analysis), S19, S20
- Ma 108 (Classical Analysis), W18, W19, W20
- Ma 191 (Selected topics), S18, W20
- Ma/CS 6 (Introduction to Discrete Mathematics), W18, W19

- Ma 098 (Individual reading course), F18

University of Bonn

- TA for Mathematics for Geoscientists, W13–14
- TA for Analysis for Computer Scientists, S13
- TA for Linear Algebra I, W12–13

Professional service

Co-organization of scientific events

- Workshop 'Reading Groups in Analysis', U. of Pennsylvania, PA, 8/2025.
- HIM trimester program 'Boolean Analysis in Computer Science', Bonn, September 9–December 18, 2024.
- Seminar 'Probability and Analysis Webinar', F20–present.
- Spring school 'Multilinear singular integrals with applications', Madison, WI, 5/2024.
- Workshop 'Analysis on the hypercube with applications to quantum computing'. American Institute of Mathematics, San Jose, CA. 6/2022.
- Seminar 'Analysis and Geometry', Chapman U., S21–S22.
- Conference 'IWOTA 2021' (local organizer), Chapman U., Orange, CA. 8/2021.
- Seminar 'Caltech/UCLA joint Analysis seminar', W18–S20.
- Conference 'Young Women in Harmonic Analysis and PDE'. Bonn, 12/2016.
- Seminar 'Basic Notions'. U. of Bonn, WT 2014–15 and ST 2015.

Community and outreach

- Postdoctoral mentor for Mario Stipčić, 7/2021–7/2023.
- Reviewer for the NSF and the Simons Foundation.
- Category Award judge at California Science and Engineering Fair. Los Angeles, CA, 4/2019.
- Talks in Math 20 – Frontiers in Mathematics (seminar to present research topics at an introductory level to first year students). Caltech, 11/2017 and 12/2018.
- Peer-reviewer for journals. Reviewed for zbMATH and MathSciNet.
- Caltech Postdoc Association representative for Mathematics, 2019–2020.

Service at Chapman

- Search committee member, F26.
- Course coordinator for Math 100/101, F26–S27.
- Member of the Faculty Senate, F25–S27.
- Math Summer Bootcamp instructor, 8/2024.
- Member of the Academic Integrity Committee, F23–S25.

Date: October 30, 2026